Safe_Volt LABS

TITLE :
DUAL-LAYER FIRE DEFENSE : PRE-FAULT ELECTRICAL MONITORING & YOLO-BASED SURVEILLANCE FOR HOSPITALS

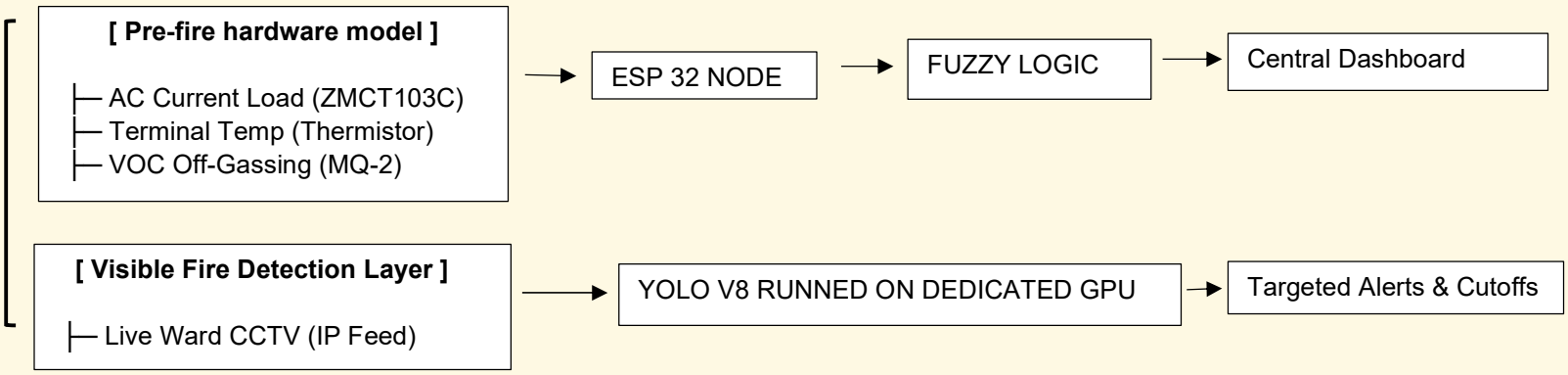
METHODOLOGY : *The Dual-Layer Architecture*

- **Layer 1**: Predictive IoT Hardware (The Invisible Threat): An ESP32 microcontroller is integrated directly into electrical distribution panels. Using multi-sensor fusion (ZMCT103C AC Current, Thermistors, and MQ-2 VOC Gas sensors), the edge node continuously evaluates localized thermal gradients against electrical load thresholds. This detects the specific heat and off-gassing signatures of melting wire insulation, predicting a fire before combustion occurs.

- **Layer 2**: AI Vision Surveillance (The Physical Threat): To protect the external corridors and wards, live IP camera feeds (simulating existing hospital CCTV) are routed to a local processing unit. A custom-trained YOLOv8 object-detection model runs on a dedicated RTX GPU to achieve zero-latency, real-time identification of open flames and smoke.

- **Central Integration**: Both the predictive hardware and the AI vision layer feed telemetry into a centralized dashboard, which triggers targeted localized alarms and automated power cutoffs to isolate faults without hospital-wide disruption.

BLOCK DIAGRAM



Proposed Prototype Outcome

A fully functional, scaled-down prototype demonstrating two distinct capabilities: successfully isolating a simulated electrical fault based on pre-fire heat and gas signatures, and identifying a live, controlled physical flame instantly using AI. The outcome proves that complex, enterprise-grade hospital safety can be achieved using accessible, low-cost engineering.